Physics-Informed Neural Modeling of Kuramoto Oscillators

From Surrogate Modeling to Hybrid and Advanced Hybrid Models for Parkinsonian Tremor Dynamics

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Abstract

Synchronization of coupled oscillators underlies phenomena ranging from power grids to neural rhythms, and the Kuramoto model is its canonical mathematical description. This work develops, in three consecutive stages, a physics-informed neural framework built around the Kuramoto system. **Part I** analyzes the Kuramoto ODE (classification, order-parameter reduction, existence by Peano's theorem, uniqueness by Osgood's criterion) and constructs a neural *surrogate* of its flow map $\Phi_T : \boldsymbol{\theta}_0 \mapsto \boldsymbol{\theta}(T)$. The surrogate is trained on Dormand–Prince RK45 data in a circular $(\cos, \sin)$ embedding with a physics-informed loss that combines mean squared error, order-parameter consistency and phase-coherence terms; on the test set it attains a mean angular error of 17.16° (0.299 rad) and an order-parameter error of 0.0080 (in a first experiment: 19.42° and 0.00264). **Part II** applies the Kuramoto model to Parkinsonian tremor recordings (ALAMEDA PD dataset: 4151 samples, 99 features). Phase trajectories are extracted by PCA and the Hilbert transform, natural frequencies and coupling are estimated from data, and a residual neural network learns the mismatch between data and physics. The resulting *hybrid* model lowers the one-step derivative MSE by 15.22%, the multi-step angular MSE by 8.57%, and reduces the relative error of the final order parameter from 61.4% to 1.0%. **Part III** extends the hybrid model by a state-dependent coupling correction $\Delta K(\boldsymbol{\theta})$, learned with the residual network frozen to keep the two roles identifiable. This *advanced hybrid* model improves the test MSE by a further 0.37% (0.13% in a 25-dimensional setting), compared with 12.06% (7.57%) gained by the residual network over pure physics, showing that most unmodeled dynamics is captured by the additive residual while the decomposition *physics + residual + coupling* remains interpretable.

Keywords: Kuramoto model, synchronization, surrogate modeling, physics-informed neural networks, hybrid modeling, residual learning, Parkinson's tremor, order parameter, Hilbert transform.

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1 Introduction and Relevance of the Problem

1.1 Why synchronization matters

Many natural and engineered systems consist of large numbers of units that oscillate and interact: generators in electrical power grids, vehicles in traffic flow, neurons and neural populations in the brain, mechanical pendulums, and even ecosystems. Whether such a collection behaves coherently or incoherently determines the stability of a power network, the throughput of a traffic system, or the presence of a pathological rhythm in the brain. The Kuramoto model [1, 5] is the standard minimal model of this collective behavior: each oscillator has its own natural frequency, the oscillators are coupled through the sine of their phase differences, and a single scalar—the *order parameter*—measures the degree of synchronization.

1.2 Why Parkinsonian tremor is a relevant application

Parkinson's disease (PD) is a neurological disorder in which tremor is one of the characteristic motor symptoms. A tremor signal is by nature oscillatory, and pathological neural oscillations in the brain are commonly modeled by coupled oscillators. Model-based studies of deep brain stimulation (DBS) predict that the optimal stimulation strategy depends on the *phase* and the *amplitude* of the ongoing tremor and that stimulation is more effective when the tremor amplitude is low; they propose hybrid closed-loop strategies combining phase-locked and adaptive DBS (see Section 9.3). Such strategies require models that can describe—and predict quickly—the instantaneous state of an oscillatory system. At the same time, wearable sensors (such as those used in the ALAMEDA project) now provide long, multivariate recordings of tremor in daily life, so that there is data from which phases, natural frequencies and couplings can be estimated.

1.3 Why neither pure physics nor pure data is enough

Two difficulties motivate this work.

(i) **Computational cost.** Solving an ODE system for many initial conditions and parameter settings can be expensive. A trained neural *surrogate* replaces the numerical integrator by a single forward pass, which is attractive for repeated evaluation, but it must respect the circular geometry of phases and the collective (synchronization) structure of the dynamics.

(ii) **Model error.** The classical Kuramoto equation is an idealization. Real tremor signals show nonlinearity, noise, asymmetric coupling and patient-specific effects that the model cannot capture, whereas purely data-driven models often lose physical structure and may diverge in long rollouts. A *hybrid* model, in the spirit of physics-informed neural networks [6], keeps the physics as a prior and lets a neural network learn only what the physics misses; the model parameters themselves can be estimated from data [7].

1.4 Objectives and structure of the report

The report consolidates three consecutive projects of the course *Advanced Topics in Numerical Mathematics*:

Part I (Project #1): **Surrogate modeling** of the Kuramoto flow map (Sections 2–8).

Part II (Project #2): **Hybrid model** (Kuramoto physics + residual neural network) for Parkinsonian tremor (Sections 9–12).

Part III (Project #3): Advanced hybrid model (physics + residual network + state-dependent coupling network) (Sections 13–15).

The report closes with a joint conclusion (Section 16); self-assessment tables of the three projects are collected in Appendix A.

Part I
Surrogate Modeling of the Kuramoto System

2 The Kuramoto Model

2.1 General description

The Kuramoto model (or Kuramoto–Daido model), first proposed by Yoshiki Kuramoto, is a mathematical model used in describing synchronization. More specifically, it is a model for the behavior of a large set of coupled oscillators. Its formulation was motivated by the behavior of systems of chemical and biological oscillators, and it has found widespread applications in areas such as neuroscience and oscillating flame dynamics. Kuramoto was quite surprised when the behavior of some physical systems, namely coupled arrays of Josephson junctions, followed his model [3, 2].

The model makes several assumptions: the coupling is weak, the oscillators are identical or nearly identical, and the interactions depend sinusoidally on the phase difference between each pair of objects.

2.2 Oscillators

An *oscillator* is a system or device that exhibits periodic motion or produces a periodic signal. Depending on the domain one distinguishes:

- **Mechanical oscillator:** a system that experiences repetitive motion, such as a pendulum or a mass–spring system.
- **Electrical oscillator:** a circuit that generates a periodic electrical signal, such as a sine-wave generator.
- **Harmonic oscillator:** a system in which the restoring force is directly proportional to the displacement, like a spring–mass system.
- **Electronic oscillator:** circuits that produce alternating current (AC) signals, often used in communications or as clocks in digital systems.

The key characteristics of an oscillator are its **amplitude** (maximum displacement from the equilibrium position), **frequency** (number of cycles completed per unit time), **period** (time taken for one complete cycle) and **phase** (the position of the oscillator within its cycle at a given time).

Typical examples are: the quartz crystal oscillator (used in watches and computers to keep time); the heart rhythm (a biological oscillator); a plucked guitar string vibrating at its natural frequency; the electrical LC circuit (inductor L and capacitor C); springs and masses following Hooke's law; neuronal oscillations (brain waves in alpha, beta and gamma bands); and planetary motion, which can be modeled as an oscillator in celestial mechanics.

2.3 Governing equations

In the most popular version of the Kuramoto model, each oscillator has its own intrinsic natural frequency ω_i, and all oscillators are equally coupled. Surprisingly, this fully nonlinear model can be solved exactly in the limit of infinitely many oscillators, $N \to \infty$: using self-consistency arguments one obtains steady-state solutions for the order parameter. The governing equation of

the model *without noise* is

$$\frac{\mathrm{d}\theta_i}{\mathrm{d}t} = \omega_i + \frac{1}{N}\sum_{j=1}^{N} K_{ij}\sin(\theta_j - \theta_i), \qquad i = 1, \ldots, N, \tag{1}$$

where the system consists of N limit-cycle oscillators and θ_i is the phase of the i-th oscillator, ω_i its natural frequency, N the total number of oscillators, K_{ij} the coupling strength between oscillators i and j, and $\sin(\theta_j - \theta_i)$ measures the phase difference. With a constant coupling $K_{ij} \equiv K$ one obtains the mean-field form used below. *With noise*, the equation reads

$$\frac{\mathrm{d}\theta_i}{\mathrm{d}t} = \omega_i + \zeta_i + \frac{K}{N}\sum_{j=1}^{N} \sin(\theta_j - \theta_i), \tag{2}$$

where ζ_i is a random noise term introducing stochastic perturbations.

2.4 Synchronization

When the coupling constant K is sufficiently strong, the oscillators synchronize to a common frequency: they adjust their phases and eventually oscillate together in unison. If K is weak, the oscillators continue to oscillate independently, without synchronization (Figure 1).

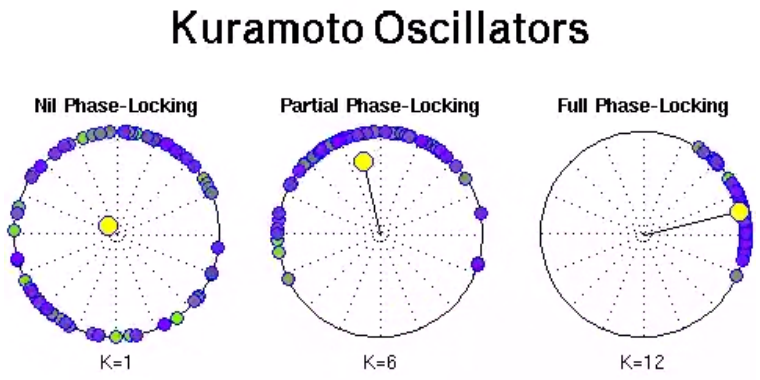


Figure 1: Kuramoto oscillators on the unit circle for increasing coupling: no phase-locking ($K = 1$), partial phase-locking ($K = 6$) and full phase-locking ($K = 12$).

2.5 Classification of the governing equation

By order. The Kuramoto model is a first-order ordinary differential equation: the highest derivative present is the first derivative $\mathrm{d}\theta_i/\mathrm{d}t$, hence the order is 1.

By linearity. The model is a *nonlinear* system. The dependent variable θ_i and its derivative appear linearly, but the interaction between the oscillators is described by the sine function, $\sin(\theta_j - \theta_i)$, which makes the phases interact nonlinearly. *Conclusion: the Kuramoto model is a nonlinear system of first-order ODEs.*

Homogeneous vs. nonhomogeneous. The model is homogeneous in the sense that there are no external forcing terms (no term $f(t)$): the evolution of each oscillator depends only on its natural frequency ω_i and the couplings K_{ij}, and the system is self-contained—its dynamics are entirely determined by internal interactions.

Constant coefficients. The natural frequencies ω_i and the couplings K_{ij} are fixed, independent of time or of other variables, so the system has constant coefficients. If $\omega_i(t)$ or $K_{ij}(t)$ changed with time or with external conditions, the coefficients would become variable.

By variable type. The *independent variable* is the time t: it dictates the progression of the system, and all dynamics evolve with respect to t. The *dependent variables* are the phases $\theta_i(t)$, $i = 1, \ldots, N$, each representing the state of the i-th oscillator at time t; they depend on t and on the interactions with the other oscillators through the coupling terms.

Table 1: Classification of the Kuramoto equation.

Criterion	Result
Order	first order
Linearity	nonlinear (sine coupling)
Forcing	homogeneous (no external forcing $f(t)$)
Coefficients	constant (ω_i, K_{ij} fixed)
Independent variable	time t
Dependent variables	phases $\theta_i(t)$, $i = 1, \ldots, N$

2.6 Order-parameter transformation

The Kuramoto model describes phase oscillators coupled with strength K that attempt to synchronize. To simplify and analyze the system one introduces the *order parameter*

$$re^{\mathrm{i}\psi} = \frac{1}{N}\sum_{j=1}^{N} e^{\mathrm{i}\theta_j}, \tag{3}$$

where r (phase coherence) measures the synchronization level and ψ (average phase) is the mean phase of the oscillators (Figure 2). The Kuramoto equation then becomes

$$\frac{\mathrm{d}\theta_i}{\mathrm{d}t} = \omega_i + Kr\sin(\psi - \theta_i). \tag{4}$$

This transformation decouples the individual oscillators, replacing direct pairwise interactions by the global parameters r and ψ.

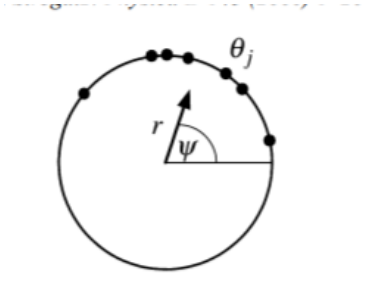


Figure 2: Geometric interpretation of the order parameter: the phases θ_j are plotted on the unit circle and their centroid is the complex number $re^{\mathrm{i}\psi}$, shown as an arrow. Source: S. H. Strogatz, *Physica D* **143** (2000) 1–20 [5].

Choosing a rotating reference frame in which $\psi = 0$ simplifies the equation to

$$\frac{\mathrm{d}\theta_i}{\mathrm{d}t} = \omega_i - Kr\sin\theta_i. \tag{5}$$

This allows the analysis of the stationary points ($\mathrm{d}\theta_i/\mathrm{d}t = 0$), of the stability of synchronized states, and of initial and boundary conditions. *Conclusion:* the order-parameter transformation simplifies the model and enables the analytical study of phase synchronization, stability and uniqueness analysis, and the formulation of boundary and initial conditions.

2.7 Stationary synchronization for mean-field coupling

The order parameter can be expressed through the phase distribution as

$$re^{\mathrm{i}\psi} = \int_{-\pi}^{\pi} e^{\mathrm{i}\theta} \Big(\frac{1}{N} \sum_{j=1}^{N} \delta(\theta - \theta_j) \Big) \mathrm{d}\theta.$$

When $N \to \infty$ the phases are described by a continuous distribution $\rho(\theta, \omega, t)$, and

$$re^{\mathrm{i}\psi} = \int_{-\pi}^{\pi} \int_{-\infty}^{\infty} e^{\mathrm{i}\theta} \rho(\theta, \omega, t)\, g(\omega)\, \mathrm{d}\theta\, \mathrm{d}\omega,$$

where $g(\omega)$ is the distribution of natural frequencies and $\rho(\theta, \omega, t)$ is the phase distribution at time t.

- For $K \to 0$ the oscillators are not synchronized: each moves with its natural frequency, $\theta_i \approx \omega_i t + \theta_i(0)$, and $r \to 0$.
- For $K \to \infty$ the oscillators synchronize and $r = 1$.
- In intermediate cases $K_c < K < \infty$ *partial synchronization* occurs: some oscillators are synchronized, others are not, and $0 < r < 1$.

Global synchronization: all oscillators move in the same phase. *Clustering*: the oscillators are partially synchronized, e.g. some move at one frequency and others at another. The phase distribution evolves according to the continuity equation

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial \theta} \Big[\big(\omega + Kr \sin(\psi - \theta) \big) \rho \Big] = 0.$$

In the incoherent state ($r = 0$) the oscillators are not synchronized. If $K < K_c$ the oscillators move incoherently ($r \to 0$); if $K > K_c$ partial synchronization occurs ($0 < r < 1$); and if $K \to \infty$ all oscillators move in the same phase ($r = 1$). The typical evolution of $r(t)$ is shown schematically in Figure 3. The Kuramoto model is thus a useful tool for analyzing synchronization, and the order parameter r represents the degree of synchronization.

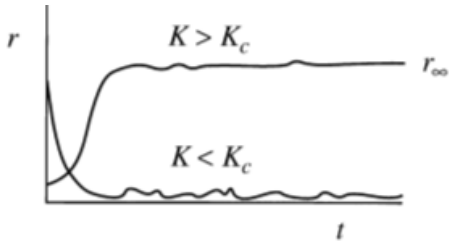


Figure 3: Schematic illustration of the typical evolution of $r(t)$ seen in numerical simulations of the Kuramoto model, for $K > K_c$ (converging to r_∞) and $K < K_c$. Source: S. H. Strogatz, *Physica D* **143** (2000) 1–20 [5].

3 Continuity, Existence and Uniqueness

The Kuramoto system can be written as

$$\frac{\mathrm{d}\theta_i}{\mathrm{d}t} = f_i(t,\boldsymbol{\theta}), \quad i = 1,\dots,N, \qquad f_i(t,\boldsymbol{\theta}) = \omega_i + \frac{K}{N}\sum_{j=1}^{N}\sin(\theta_j - \theta_i). \tag{6}$$

3.1 Continuity of the vector field

To show that $\mathbf{f}(t,\boldsymbol{\theta})$ is continuous, consider a sequence $\boldsymbol{\theta}^{(n)} = (\theta_1^{(n)},\dots,\theta_N^{(n)})$ converging to $\boldsymbol{\theta} = (\theta_1,\dots,\theta_N)$; we must show $\mathbf{f}(t,\boldsymbol{\theta}^{(n)}) \to \mathbf{f}(t,\boldsymbol{\theta})$. Since the sine function is continuous,

$$\theta_j^{(n)} \to \theta_j \;\Rightarrow\; \sin(\theta_j^{(n)} - \theta_i^{(n)}) \to \sin(\theta_j - \theta_i).$$

Each term of the sum converges pointwise and the sum is finite (N terms), hence

$$\sum_{j=1}^{N}\sin(\theta_j^{(n)} - \theta_i^{(n)}) \to \sum_{j=1}^{N}\sin(\theta_j - \theta_i),$$

and therefore

$$f_i(t,\boldsymbol{\theta}^{(n)}) = \omega_i + \frac{K}{N}\sum_{j=1}^{N}\sin(\theta_j^{(n)} - \theta_i^{(n)}) \to \omega_i + \frac{K}{N}\sum_{j=1}^{N}\sin(\theta_j - \theta_i) = f_i(t,\boldsymbol{\theta}).$$

Since every component converges, $\mathbf{f}(t,\boldsymbol{\theta}^{(n)}) \to \mathbf{f}(t,\boldsymbol{\theta})$, i.e. the vector field is continuous.

3.2 Existence of solutions

Continuity allows us to apply existence theorems such as Peano's.

Theorem (Peano's existence theorem)**.** *Consider the initial value problem* $\dfrac{\mathrm{d}y}{\mathrm{d}t} = f(t,y)$, $y(t_0) = y_0$, *where* $f : D \subseteq \mathbb{R}\times\mathbb{R}^n \to \mathbb{R}^n$ *is continuous on a domain* D. *Then there exists a solution* $y(t)$ *on some interval* $[t_0-\alpha, t_0+\alpha]$, $\alpha > 0$.

3.3 Uniqueness of the solution

We prove uniqueness by the Osgood uniqueness criterion.

Theorem (Osgood criterion)**.** *Suppose there is a continuous, non-decreasing function* $\varpi : [0,\infty) \to [0,\infty)$ *such that (1)* $\varpi(0) = 0$, *(2)* $\varpi(u) > 0$ *for* $u > 0$, *(3)* $\int_0^{\varepsilon} \frac{\mathrm{d}u}{\varpi(u)} = \infty$ *for some* $\varepsilon > 0$, *and (4)* $\|\mathbf{f}(t,\boldsymbol{\theta}_1) - \mathbf{f}(t,\boldsymbol{\theta}_2)\| \le \varpi(\|\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2\|)$ *for all* $(\boldsymbol{\theta}_1,\boldsymbol{\theta}_2)$ *in the domain. Then the solution of the initial value problem is unique.*

For each component i,

$$|f_i(t,\boldsymbol{\theta}_1) - f_i(t,\boldsymbol{\theta}_2)| = \frac{K}{N}\Big|\sum_{j=1}^{N}\Big(\sin(\theta_j^{(1)} - \theta_i^{(1)}) - \sin(\theta_j^{(2)} - \theta_i^{(2)})\Big)\Big|.$$

By the mean value theorem for the sine there is $\xi_{j,i}$ between $\theta_j^{(1)} - \theta_i^{(1)}$ and $\theta_j^{(2)} - \theta_i^{(2)}$ with

$$\sin(\theta_j^{(1)} - \theta_i^{(1)}) - \sin(\theta_j^{(2)} - \theta_i^{(2)}) = \cos(\xi_{j,i})\big[(\theta_j^{(1)} - \theta_i^{(1)}) - (\theta_j^{(2)} - \theta_i^{(2)})\big].$$

Since $|\cos \xi_{j,i}| \leq 1$,

$$\left|\sin(\theta_j^{(1)} - \theta_i^{(1)}) - \sin(\theta_j^{(2)} - \theta_i^{(2)})\right| \leq |\theta_j^{(1)} - \theta_j^{(2)}| + |\theta_i^{(1)} - \theta_i^{(2)}|.$$

Summing over j and multiplying by K/N,

$$|f_i(t, \boldsymbol{\theta}_1) - f_i(t, \boldsymbol{\theta}_2)| \leq \frac{K}{N}\Big(\sum_{j=1}^{N} |\theta_j^{(1)} - \theta_j^{(2)}| + N|\theta_i^{(1)} - \theta_i^{(2)}|\Big).$$

Taking the norm over i yields a Lipschitz estimate

$$\|\mathbf{f}(t, \boldsymbol{\theta}_1) - \mathbf{f}(t, \boldsymbol{\theta}_2)\| \leq L\,\|\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2\|, \qquad L = 2K. \tag{7}$$

Hence we may choose $\varpi(u) = Lu$, and

$$\int_0^{\varepsilon} \frac{\mathrm{d}u}{\varpi(u)} = \frac{1}{L}\int_0^{\varepsilon} \frac{\mathrm{d}u}{u} = \infty,$$

because $\int_0^{\varepsilon} \mathrm{d}u/u$ diverges. Since $\mathbf{f}$ satisfies the conditions of the Osgood criterion, the solution of the Kuramoto differential equation is unique.

4 Practical Applications of the Kuramoto Model

The Kuramoto model is widely applied because of its ability to describe synchronization phenomena. Key examples are:

1. **Power grid analysis.** The model is used to study the synchronization of generators within electrical power grids. It helps in understanding how generators synchronize their phases and frequencies, which is crucial for the stability and reliability of power distribution systems.
2. **Traffic flow management.** In traffic networks the model is applied to model and control the synchronization of vehicle speeds. Treating traffic waves as coupled oscillators facilitates strategies that improve traffic flow and reduce congestion at intersections.
3. **Neuroscience.** The model provides insights into the synchronization of neural oscillations in the brain, which is essential for understanding cognitive functions and neurological conditions.
4. **Mechanical systems.** The model has been applied to synchronize mechanical systems such as pendulums. By coupling pendulums with appropriate parameters, researchers can study and control their synchronization behavior, with implications for designing coordinated mechanical systems.
5. **Ecological systems.** The model has been employed to explore synchronization phenomena in ecological systems and offers a novel perspective on the structural dynamics of ecosystems.

5 Numerical Method for Data Generation

Training data for the surrogate are generated by high-accuracy numerical integration of the Kuramoto system. We use an explicit Runge–Kutta method with adaptive step size (Dormand–Prince RK45).

5.1 Initial value problem

We consider the initial value problem

$$\dot{y}(t) = f(t, y(t)), \qquad y(t_0) = y_0,$$

where $y(t) \in \mathbb{R}^d$ is the state vector and $f : \mathbb{R} \times \mathbb{R}^d \to \mathbb{R}^d$ defines the system dynamics. The solution is approximated on a discrete time grid $t_{n+1} = t_n + h$, $h > 0$.

5.2 General explicit Runge–Kutta scheme

An explicit s-stage Runge–Kutta method computes intermediate slopes

$$\begin{aligned}
k_1 &= f(t_n, y_n), \\
k_2 &= f(t_n + c_2 h,\ y_n + h a_{21} k_1), \\
k_3 &= f(t_n + c_3 h,\ y_n + h(a_{31} k_1 + a_{32} k_2)), \\
&\vdots \\
k_s &= f\Big(t_n + c_s h,\ y_n + h \sum_{j=1}^{s-1} a_{sj} k_j\Big),
\end{aligned}$$

and the solution is updated as

$$y_{n+1} = y_n + h \sum_{i=1}^{s} b_i k_i.$$

The method is *explicit* because $a_{ij} = 0$ for all $j \geq i$. This implies that each stage k_i depends only on previously computed stages, that no nonlinear system has to be solved at each time step, and that the method is computationally efficient.

5.3 Dormand–Prince RK45 (embedded method)

The Dormand–Prince RK45 method is an embedded Runge–Kutta scheme that simultaneously computes a fourth-order approximation $y_{n+1}^{(4)}$ and a fifth-order approximation $y_{n+1}^{(5)}$. Both use the same intermediate slopes k_i:

$$y_{n+1}^{(p)} = y_n + h \sum_{i=1}^{s} b_i^{(p)} k_i, \qquad p = 4, 5,$$

so that no additional function evaluations are required.

5.4 Local error estimation and adaptive step size

The local truncation error is estimated by comparing the two embedded solutions,

$$\mathrm{err}_n = \big\| y_{n+1}^{(5)} - y_{n+1}^{(4)} \big\|,$$

which gives a reliable measure of the numerical error introduced in one step without additional function evaluations. The estimated error is compared against the user-defined tolerance

$$\mathrm{tol} = \mathrm{ATOL} + \mathrm{RTOL} \cdot \|y_n\|.$$

If $\mathrm{err}_n > \mathrm{tol}$, the step is rejected and the step size is reduced; if $\mathrm{err}_n \leq \mathrm{tol}$, the step is accepted and the solution is updated. The step size is adjusted by

$$h_{\mathrm{new}} = h \left(\frac{\mathrm{tol}}{\mathrm{err}_n} \right)^{\frac{1}{p+1}}, \qquad p = 4,$$

which balances accuracy and computational efficiency.

6 General Idea of Surrogate Modeling for ODEs

Many dynamical systems are defined by an ODE

$$\frac{\mathrm{d}f}{\mathrm{d}t} = A(f,t), \qquad f(0) = f_0.$$

Solving the ODE for many different initial conditions can be computationally expensive. The *surrogate idea* is to replace the ODE solver by a neural network that approximates its output; the network learns the mapping

$$f_0 \mapsto f(t_{\max}) \qquad \text{or} \qquad (t, f_0) \mapsto f(t).$$

Step 1: data generation. Choose a set of initial conditions $f_0^{(i)}$; for each, solve the ODE with a high-accuracy solver (e.g. RK45, BDF) and record the outputs $\big(f_0^{(i)}, f^{(i)}(t_{\max})\big)$ or $\big(f_0^{(i)}, t_j, f^{(i)}(t_j)\big)$. These pairs form the training dataset of the surrogate.

Step 2: dataset structure. Two common formats are (1) *end-point mapping* $\big(f_0^{(i)}, f^{(i)}(t_{\max})\big)$ and (2) *full-trajectory mapping* $\big(f_0^{(i)}, t_j\big) \mapsto f^{(i)}(t_j)$. The second approach learns a continuous surrogate of the dynamics; data are usually sampled on a time grid $t_0, t_1, \ldots, t_{\max}$.

Step 3: training. Choose a model architecture (an MLP is typical) and solve

$$\min_{\theta} \ \mathrm{MSE}\big(\mathrm{NN}_\theta(\text{input}),\ \text{true ODE solution}\big).$$

The neural network learns to approximate the ODE dynamics without explicitly solving differential equations. After training, prediction is extremely fast, since no numerical integrator is needed.

7 Surrogate Modeling of the Kuramoto System

7.1 Problem statement

Given N coupled phase oscillators with state $\boldsymbol{\theta}(t) = (\theta_1(t), \ldots, \theta_N(t)) \in \mathbb{R}^N$ and Kuramoto dynamics

$$\dot{\theta}_i(t) = \omega_i + \frac{K}{N} \sum_{j=1}^{N} \sin(\theta_j(t) - \theta_i(t)), \qquad i = 1, \ldots, N,$$

with initial condition $\boldsymbol{\theta}(0) = \boldsymbol{\theta}_0$ and known K and $\boldsymbol{\omega}$. The superscript (k) denotes the k-th sample of the dataset, i.e. $\boldsymbol{\theta}_0^{(k)}$ is the k-th randomly generated initial condition. The **goal** is to learn a fast surrogate that approximates the flow map

$$\Phi_T : \boldsymbol{\theta}_0 \mapsto \boldsymbol{\theta}(T)$$

without running an ODE solver each time.

7.2 Surrogate method

1. Generate a dataset by numerical integration: $\boldsymbol{\theta}_T^{(k)} = \Phi_T(\boldsymbol{\theta}_0^{(k)})$, $k = 1, \ldots, M$.
2. Convert phases to a continuous form to avoid 2π wrapping: $c(\boldsymbol{\theta}) = (\cos\boldsymbol{\theta}, \sin\boldsymbol{\theta}) \in \mathbb{R}^{2N}$.
3. Train a neural network $f_w : \mathbb{R}^{2N} \to \mathbb{R}^{2N}$ such that $f_w(c(\boldsymbol{\theta}_0)) \approx c(\boldsymbol{\theta}_T)$.

7.3 Data and representation design

Data generation. We create M pairs $\{(\boldsymbol{\theta}_0^{(k)}, \boldsymbol{\theta}_T^{(k)})\}_{k=1}^M$ with $\boldsymbol{\theta}_T^{(k)} = \Phi_T(\boldsymbol{\theta}_0^{(k)}; \boldsymbol{\omega}^{(k)}, K)$, where $\boldsymbol{\theta}_0^{(k)}$ and (possibly) $\boldsymbol{\omega}^{(k)}$ are randomly generated samples (e.g. $\boldsymbol{\theta}_0 \sim \mathcal{U}(-\pi, \pi)^N$, $\boldsymbol{\omega} \sim \mathcal{N}(0, \sigma_\omega^2 I)$) and Φ_T is computed with `solve_ivp`.

Circular representation. Phases are elements of a circle and differences should be taken modulo 2π. Therefore we apply the transformation $c(\boldsymbol{\theta}) := (\cos\boldsymbol{\theta}, \sin\boldsymbol{\theta}) \in \mathbb{R}^{2N}$, so that each example becomes $X^{(k)} = c(\boldsymbol{\theta}_0^{(k)})$, $Y^{(k)} = c(\boldsymbol{\theta}_T^{(k)})$.

7.4 Parametrization and minimization

We choose a neural network $f_w : \mathbb{R}^{2N} \to \mathbb{R}^{2N}$ with parameters w whose goal is $f_w(c(\boldsymbol{\theta}_0)) \approx c(\Phi_T(\boldsymbol{\theta}_0))$, and minimize

$$\min_w \ \frac{1}{M} \sum_{k=1}^{M} \mathcal{L}\big(f_w(X^{(k)}), Y^{(k)}\big),$$

where M is the number of training samples and $\mathcal{L}$ denotes the physics-informed loss.

7.5 Physics-informed loss

MSE on circular coordinates.

$$\mathcal{L}_{\text{MSE}}(w) = \frac{1}{M} \sum_{k=1}^{M} \big\| f_w(X^{(k)}) - Y^{(k)} \big\|_2^2,$$

where the norm is taken in $\mathbb{R}^{2N}$ (cos/sin data).

Order-parameter loss. The order parameter of the Kuramoto model is

$$r(\boldsymbol{\theta}) = \Big| \frac{1}{N} \sum_{j=1}^{N} e^{\mathrm{i}\theta_j} \Big| = \sqrt{\Big(\frac{1}{N} \sum_j \cos\theta_j \Big)^2 + \Big(\frac{1}{N} \sum_j \sin\theta_j \Big)^2},$$

and the loss term is

$$\mathcal{L}_r(w) = \frac{1}{M} \sum_{k=1}^{M} \Big(r(\hat{c}^{(k)}) - r(c^{(k)}) \Big)^2,$$

where $\hat{c}^{(k)} = f_w(X^{(k)})$ and $c^{(k)} = Y^{(k)}$.

Phase-coherence (variance) loss. We compute the statistical variance of the phases (a precision proxy) and compare:

$$\mathcal{L}_{\text{phase}}(w) = \frac{1}{M} \sum_{k=1}^{M} \Big(\operatorname{Var}(\hat{\boldsymbol{\theta}}^{(k)}) - \operatorname{Var}(\boldsymbol{\theta}^{(k)}) \Big)^2,$$

where $\hat{\boldsymbol{\theta}}^{(k)} = \operatorname{atan2}\big(\hat{c}_{\sin}^{(k)}, \hat{c}_{\cos}^{(k)}\big)$ maps the predicted sine and cosine values to phases in $(-\pi, \pi]$, ensuring correct angle reconstruction and preservation of the phase-distribution structure. $\operatorname{Var}(\boldsymbol{\theta})$ measures how spread out the phases are; the difference $\operatorname{Var}(\hat{\boldsymbol{\theta}}^{(k)}) - \operatorname{Var}(\boldsymbol{\theta}^{(k)})$ penalizes deviations between predicted and true phase dispersion and thus encourages correct phase coherence.

Total loss.

$$\mathcal{L}(w) = \alpha_{\text{MSE}}\mathcal{L}_{\text{MSE}} + \alpha_r\mathcal{L}_r + \alpha_{\text{phase}}\mathcal{L}_{\text{phase}}, \tag{8}$$

where $\alpha_{\text{MSE}}, \alpha_r, \alpha_{\text{phase}}$ are hyperparameters (weights) balancing all components.

7.6 Optimization and training

Gradient-based update (AdamW). Standard gradient descent updates $w \leftarrow w - \eta\nabla_w\mathcal{L}(w)$, where w denotes the network parameters (weights and biases), η is the learning rate controlling the step size, and $\nabla_w\mathcal{L}(w)$ is the gradient of the loss with respect to w. AdamW is an adaptive optimization method that improves convergence by using first- and second-order moment estimates of the gradients.

Mini-batch optimization. Instead of the full dataset, training is performed on mini-batches,

$$\mathcal{L}_{\text{batch}}(w) = \frac{1}{|B|}\sum_{k\in B}\mathcal{L}\big(f_w(X^{(k)}), Y^{(k)}\big),$$

where B is a subset (mini-batch) of training samples and $|B|$ its size; the gradients $\nabla_w\mathcal{L}_{\text{batch}}$ are used for the parameter updates.

7.7 Evaluation metrics

Angular distance on the circle. Phase differences must respect periodicity:

$$\text{wrap}(x) = \big((x+\pi) \bmod 2\pi\big) - \pi, \qquad d(\hat{\theta}, \theta) = \big|\,\text{wrap}(\hat{\theta} - \theta)\big|,$$

where wrap maps angular differences to the interval $(-\pi, \pi]$, ensuring the shortest distance on the circle.

Angular error metrics.

$$\text{MAE}_\theta = \frac{1}{MN}\sum_{k=1}^{M}\sum_{i=1}^{N} d\big(\hat{\theta}_i^{(k)}, \theta_i^{(k)}\big), \qquad \text{RMSE}_\theta = \sqrt{\frac{1}{MN}\sum_{k=1}^{M}\sum_{i=1}^{N} d\big(\hat{\theta}_i^{(k)}, \theta_i^{(k)}\big)^2},$$

where M is the number of test samples, N the number of oscillators, i indexes oscillators and k indexes samples.

Order-parameter error.

$$\text{MAE}_r = \frac{1}{M}\sum_{k=1}^{M}\Big|r(\hat{c}^{(k)}) - r(c^{(k)})\Big|,$$

which measures how well the surrogate preserves the global synchronization.

Why multiple metrics? Angular MAE/RMSE evaluate local phase accuracy, whereas MAE_r evaluates the global collective behavior. A target criterion (example) is $\text{MAE}_r < 0.01$.

7.8 Final prediction formula

$$\hat{\boldsymbol{\theta}}_T(\boldsymbol{\theta}_0) = \operatorname{atan2}\Big([f_{w^*}(c(\boldsymbol{\theta}_0))]_{\sin},\, [f_{w^*}(c(\boldsymbol{\theta}_0))]_{\cos}\Big) \approx \Phi_T(\boldsymbol{\theta}_0),$$

where $\boldsymbol{\theta}_0$ is the initial phase configuration, $c(\boldsymbol{\theta}_0) = (\cos\boldsymbol{\theta}_0, \sin\boldsymbol{\theta}_0)$ the circular embedding, f_{w^*} the trained surrogate, $[\cdot]_{\sin}, [\cdot]_{\cos}$ the predicted sine and cosine components, and atan2 reconstructs phases in $(-\pi, \pi]$. The neural network directly approximates the flow map $\Phi_T : \boldsymbol{\theta}_0 \mapsto \boldsymbol{\theta}(T)$, and a single forward pass replaces numerical ODE integration. *Key property:* fast inference with physics-informed training that preserves global synchronization and phase dispersion.

8 Experiments and Results of the Surrogate Model

Two experimental configurations were run; they are reported in the order in which they were carried out.

8.1 Experiment A

Table 2: Experiment A: problem setup and training configuration.

System parameters	**Dataset and representation**
Number of oscillators: $N = 10$	Dataset size: 10,000 samples (train/val/test = 8000/1000/1000)
Coupling strength: $K = 2.0$	Input: $(\cos\boldsymbol{\theta}_0, \sin\boldsymbol{\theta}_0) \in \mathbb{R}^{20}$
Natural frequencies ω_i: randomly generated once, then fixed	Output: $(\cos\boldsymbol{\theta}_T, \sin\boldsymbol{\theta}_T) \in \mathbb{R}^{20}$
Initial phases: uniform random in $[0, 2\pi]$	**Loss function:** MSE, order-parameter preservation, coherence measure, complex consistency
Final time: $T = 3$	**Architecture:** hidden layers $[768, 512, 512, 256, 128]$
Numerical integration: `solve_ivp` (RK45), atol = rtol = 10^{-9}	Batch size 256, learning rate 0.0003; weights $\alpha_{\text{MSE}} = 1.0$, $\alpha_{\text{order}} = 0.5$, $\alpha_{\text{phase}} = 0.3$

Table 3: Experiment A: training progress.

Epoch	Train loss	Val loss	LR
1	1.703	0.529	3.00×10^{-4}
20	0.116	0.108	3.00×10^{-4}
100	0.062	0.101	3.00×10^{-4}
160	0.035	0.102	1.50×10^{-4}
240	0.025	0.105	3.75×10^{-5}

Early stopping occurred at epoch 251 (best: epoch 101, loss 9.25×10^{-2}).

Final results (test set). The angular errors are MAE = 0.3389 rad = 19.42°, RMSE = 0.4927 rad = 28.23° and maximum error 3.133 rad = 179.51°. For the order parameter, MAE = 0.00264 and RMSE = 0.01364.

Interpretation. The surrogate achieves MAE = 19.42° and RMSE = 28.23°, indicating that it predicts oscillator phases with reasonable accuracy; large maximum errors are rare and correspond to outlier cases. The order-parameter MAE of 0.00264 shows that the global synchronization of the system is well preserved, i.e. the surrogate respects the collective dynamics of the oscillators, while producing fast predictions. *Conclusion:* the surrogate captures both local phase behavior and global order and offers an efficient alternative to full numerical integration for Kuramoto systems.

8.2 Experiment B

Surrogate model and training. The network is a fully connected feedforward network with layers $[20 \to 128 \to 128 \to 20]$ and Tanh activation. Training uses the Adam optimizer (learning rate 10^{-3}), batch size 256, at most 150 epochs and early stopping with patience 20. The loss is $\mathcal{L} = \alpha_{\text{MSE}}\mathcal{L}_{\text{MSE}} + \alpha_r\mathcal{L}_r + \alpha_{\text{phase}}\mathcal{L}_{\text{phase}}$ with $\alpha_r = 0.5$, $\alpha_{\text{phase}} = 0.3$.

Table 4: Experiment B: problem setup.

Kuramoto system parameters	**Dataset and representation**
Number of oscillators: $N = 10$	Training samples: 5000; test samples: 500
Coupling strength: $K = 2.0$	Input: $(\cos\boldsymbol{\theta}_0, \sin\boldsymbol{\theta}_0) \in \mathbb{R}^{20}$
Frequency spread: $\sigma_\omega = 0.8$	Output: $(\cos\boldsymbol{\theta}_T, \sin\boldsymbol{\theta}_T) \in \mathbb{R}^{20}$
Initial phases: $\theta_i(0) \sim \mathcal{U}(-\pi, \pi)$	**Physics-informed loss:** MSE, order-parameter consistency, phase-variance preservation
Final time: $T = 5.0$	
Numerical integration: `solve_ivp` (RK45), 50 stored trajectory points, atol = rtol = 10^{-8}	

Table 5: Experiment B: training progress.

Epoch	Total loss	MSE	L_r
5	0.720	0.528	0.319
25	0.214	0.137	0.146
50	0.165	0.111	0.102
100	0.089	0.078	0.016
150	0.083	0.075	0.010

The convergence is stable without oscillations, the physics-based losses decrease consistently, and the best loss achieved is 0.0830.

Final results (test set). $\text{MAE}_\theta = 0.299$ rad (17.16°), $\text{RMSE}_\theta = 0.417$ rad, and $\text{MAE}_r = 0.0080$; the synchronization structure is preserved.

Interpretation. *Phase accuracy:* the surrogate predicts individual oscillator phases with moderate error; large angular deviations occur rarely and correspond to isolated outliers. *Collective dynamics:* the extremely low MAE of the order parameter (0.008) indicates that the global synchronization behavior is accurately preserved. *Physical consistency:* incorporating the order-parameter and phase-variance losses significantly improves the model's ability to respect the underlying Kuramoto physics. *Conclusion:* the physics-informed surrogate captures both local phase dynamics and global collective behavior, providing a fast and reliable alternative to repeated numerical integration.

8.3 Visual representation

Figure 4 compares the initial phases, the final phases obtained by the ODE solver at $t = 5.0\,\mathrm{s}$ and the final phases predicted by the surrogate; Figure 5 shows the evolution of the order parameter $r(t)$ over time.

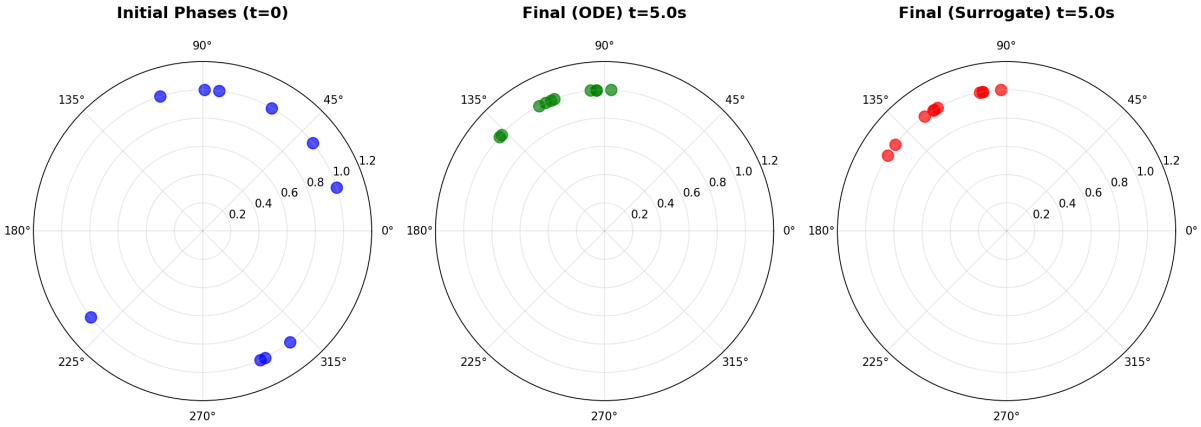


Figure 4: Phases of the oscillators on the unit circle: initial phases ($t = 0$), final phases from the ODE solver ($t = 5.0\,\mathrm{s}$) and final phases predicted by the surrogate ($t = 5.0\,\mathrm{s}$).



Figure 5: Order-parameter evolution $r(t)$ over time, $t \in [0, 5]\,\mathrm{s}$.

Part II

Hybrid Model: Kuramoto Physics with a Residual Neural Network

9 Problem Setting and Data

9.1 Problem setting and goal

The **goal** is to model Parkinson's tremor dynamics using coupled oscillators inspired by the Kuramoto model. The data come from the *ALAMEDA PD tremor dataset*, which contains time-series recordings of hand tremors in Parkinson's patients. Each tremor signal reflects oscillatory behavior and can be interpreted as the phase trajectory $\theta_i(t)$ of an oscillator.

9.2 Objective

1. **Extract phase trajectories** $\theta_i(t)$ from the tremor signals using the Hilbert transform.
2. **Estimate parameters:** the natural frequencies ω_i and the coupling constant K.
3. **Compute residual dynamics:** the difference between the observed data and the classical Kuramoto model.
4. **Train a neural-network correction term** f_{NN} that learns the residual dynamics (physics-informed approach).
5. **Evaluate performance:** compare predicted and true phase trajectories and evaluate the synchronization (order parameter).

9.3 Deep brain stimulation and phase–amplitude effects

Model-based prediction. Deep brain stimulation (DBS) can be an effective treatment for essential tremor (ET). Using a mathematical model based on coupled oscillators, it is predicted that for some ET patients the optimal stimulation strategy depends on (i) the **phase** of the ongoing tremor and (ii) the **amplitude** of the tremor oscillations. The model further predicts that stimulation effects are **stronger when the tremor amplitude is low**.

Scientific insight and clinical evidence. A reduced Kuramoto-type model is used to represent pathological neural oscillations. The analysis shows that the DBS response depends on the instantaneous state of the oscillatory system (phase and amplitude), and clinical data from ET patients support these predictions. A new **hybrid closed-loop DBS strategy** is proposed, combining phase-locked DBS and adaptive DBS [8, 9].

9.4 Dataset and problem context

ALAMEDA PD tremor dataset. The dataset comes from the ALAMEDA project, designed to monitor Parkinson's disease (PD) symptoms using wearable sensors. It contains data from **11 PD patients**, each identified by a unique `subject_id`. Each record includes hundreds of time-dependent features (e.g. acceleration, rotation, amplitude statistics) extracted from wrist or pocket sensors during daily activities. These features are sampled sequentially over time, therefore the data form a **multivariate time series**.

What we model. Tremor signals exhibit oscillatory behavior, similar to coupled neural or muscular oscillators. For each patient, sensor dynamics can be mapped to a phase variable $\theta_i(t)$ describing rhythmic motion. The aim is to model these oscillations and their interaction across patients.

9.5 Dataset description

The ALAMEDA PD tremor dataset is a sensor-based dataset related to Parkinson's disease tremor analysis. It contains **4151 samples** and **99 features** extracted from motion sensors placed on the body. The dataset includes:

- time markers (start/end timestamp, subject ID);
- amplitude-based statistical features (mean, standard deviation, variance, median, IQR);
- frequency-domain (FFT) features capturing dominant frequency, power and spectral characteristics;
- PCA-based features (PC1 features);
- clinical labels: rest tremor, kinetic tremor, postural tremor, constancy of rest.

9.6 Research objective — what are we looking for?

Each patient's tremor signal is transformed into a **phase trajectory** $\theta_i(t)$. From these phases we estimate the natural frequencies ω_i (the intrinsic tremor rates of each oscillator) and the coupling strength K (how strongly the oscillators influence each other). Using the Kuramoto model

$$\frac{\mathrm{d}\theta_i}{\mathrm{d}t} = \omega_i + \frac{K}{N}\sum_{j=1}^{N}\sin(\theta_j - \theta_i)$$

we predict how these phases evolve in time. The goal of the model is to compare the *real phase derivatives* (from data) with the Kuramoto-model prediction.

10 Neural-Network Embedding into the Kuramoto Model

The classical Kuramoto model is

$$\frac{\mathrm{d}\theta_i}{\mathrm{d}t} = \omega_i + \frac{K}{N}\sum_{j=1}^{N}\sin(\theta_j - \theta_i).$$

We introduce a neural-network term to capture unmodeled dynamics:

$$\frac{\mathrm{d}\theta_i}{\mathrm{d}t} = \omega_i + \frac{K}{N}\sum_{j=1}^{N}\sin(\theta_j - \theta_i) + f_{\mathrm{NN}}(\mathbf{x}_i; \mathbf{W}), \tag{9}$$

where θ_i is the phase of oscillator i, ω_i its natural frequency, K the coupling strength, f_{NN} the neural-network correction term and $\mathbf{W}$ the trainable parameters.

10.1 Why add a neural network?

Goal. Improve the classical Kuramoto model so that it can accurately reproduce real Parkinson's tremor dynamics.

Problem. The standard Kuramoto ODE cannot fully capture the complexity of real tremor signals (nonlinearity, noise, asymmetric coupling, patient-specific effects).

Idea. Augment the physics model with a neural-network correction term:

$$\frac{\mathrm{d}\theta_i}{\mathrm{d}t} \approx \underbrace{\dot{\theta}_i^{\text{Kuramoto}}}_{\text{known physics}} + \underbrace{f_{\text{NN}}(\boldsymbol{\theta})}_{\text{learned residual}} \quad .$$

What we seek. To learn the mismatch between the real data and the Kuramoto prediction, i.e. to capture hidden dynamics that the physical model cannot explain.

11 Methodology of the Hybrid Model

11.1 From raw data to phase trajectories

The raw tremor signals are transformed into a reduced set of phase trajectories for modeling.

1. **Raw time series / features:** $X(t) \in \mathbb{R}^{T\times\text{channels}}$ (accelerometer, gyroscope, amplitude statistics, etc.).
2. **Dimensionality reduction via PCA:** $Y(t) = X(t)W \in \mathbb{R}^{T\times N_{\text{osc}}}$ with $N_{\text{osc}} = 5$; this captures the principal modes of the oscillatory dynamics and reduces noise and redundancy.
3. **Phase extraction:** $z_i(t) = y_i(t) + \mathrm{i}\,\mathcal{H}[y_i(t)]$, $\theta_i(t) = \arg(z_i(t))$, $i = 1, \ldots, 5$; the Hilbert transform gives the analytic signal, and the phases are unwrapped/smoothed for continuity.
4. **Result:** the phase matrix.

11.2 Step 1 — Extract phase from tremor signals

Goal. Convert raw tremor measurements into a phase variable $\theta_i(t)$ suitable for oscillator modeling; Kuramoto-type dynamics operate on phases, not on raw accelerations.

- Start from the tremor time series $x_i(t)$ (e.g. accelerometer amplitude).
- Compute the analytic signal with the Hilbert transform: $z_i(t) = x_i(t) + \mathrm{i}\,\mathcal{H}[x_i(t)]$. Here $\mathcal{H}[\cdot]$ denotes the Hilbert transform, which produces a 90° phase-shifted version of the signal and allows the construction of a complex-valued analytic signal.
- Extract the instantaneous phase $\theta_i(t) = \arg(z_i(t))$; the operator $\arg(\cdot)$ returns the angle (phase) of a complex number in the complex plane.
- Smooth or unwrap the phase to ensure continuity.

Output. A clean phase trajectory $\theta_i(t)$ for each patient, ready for parameter estimation and dynamical modeling.

11.3 Step 2 — Estimate natural frequencies and coupling

Goal. Estimate the intrinsic oscillator properties for each patient.

1. **Natural frequencies ω_i.** Use the unwrapped phase $\theta_i(t)$, compute the time derivative $\dot{\theta}_i(t) \approx \mathrm{d}\theta_i/\mathrm{d}t$ and take the median over time as a robust estimate: $\omega_i = \text{median}(\dot{\theta}_i(t))$.

2. **Coupling strength K.** Use the classical Kuramoto model (the factor $1/N$ is absorbed into K)

$$\dot{\theta}_i = \omega_i + K \sum_j \sin(\theta_j - \theta_i)$$

and fit K to minimize the mismatch between $\dot{\theta}_i^{\text{data}}$ and $\dot{\theta}_i^{\text{Kuramoto}}$.

This yields ω_i and K for the baseline physics model.

11.4 Step 3 — Compute residual dynamics

Goal. Quantify what the classical Kuramoto model fails to capture. Compute the phase derivative from data, $\dot{\theta}_i^{\text{data}}$, the classical Kuramoto prediction $\dot{\theta}_i^{\text{Kuramoto}} = \omega_i + K \sum_j \sin(\theta_j - \theta_i)$, and the residual (unknown dynamics)

$$r_i(t) = \dot{\theta}_i^{\text{data}} - \dot{\theta}_i^{\text{Kuramoto}}. \tag{10}$$

The residuals $r_i(t)$ serve as targets for the neural-network training.

11.5 Step 4 — Train the neural-network correction term

Goal. Learn the unknown dynamics f_{NN} from the residuals. The model is

$$\dot{\theta}_i = \underbrace{\omega_i + K \sum_j \sin(\theta_j - \theta_i)}_{\text{known physics}} + \underbrace{f_{\text{NN}}(\theta_1, \ldots, \theta_N)}_{\text{learned residual}}.$$

Training procedure. Input: the phase vector $\boldsymbol{\theta}(t)$ (or sin/cos features); target: the residual $r_i(t)$; loss function: mean squared error

$$L = \sum_{i,t} \big(r_i(t) - f_{\text{NN}}(\boldsymbol{\theta}(t))\big)^2;$$

optimizer: Adam, with early stopping based on the validation loss. **Output:** a trained network f_{NN} that corrects the Kuramoto dynamics.

Step 4b — Neural-network training in detail

Goal. Teach f_{NN} to approximate the residual dynamics.

1. **Input features:** $X(t) = [\sin\theta_1, \ldots, \sin\theta_N, \cos\theta_1, \ldots, \cos\theta_N, \omega_1, \ldots, \omega_N]$; this encodes the current phase state of all oscillators and includes the natural frequencies to give the network temporal-scaling information.
2. **Target output:** $y(t) = r_i(t) = \dot{\theta}_i^{\text{data}} - \dot{\theta}_i^{\text{Kuramoto}}$, the residual between the real derivative and the classical Kuramoto prediction.
3. **Loss function (MSE):**

$$L = \frac{1}{NT} \sum_{i=1}^{N} \sum_{t=1}^{T} \big(y_i(t) - f_{\text{NN}}(X(t))_i\big)^2.$$

11.6 Neural-network architecture (residual learning)

The objective of the network is $r(t) = \dot{\boldsymbol{\theta}}_{\text{true}}(t) - f_{\text{physics}}(\boldsymbol{\theta}(t))$. The input features are $x = [\sin\boldsymbol{\theta}, \cos\boldsymbol{\theta}, \boldsymbol{\omega}] \in \mathbb{R}^{3N}$ and the output is $f_{\text{NN}}(x) = r(t) \in \mathbb{R}^N$. The architecture is a fully connected feedforward network with 2 hidden layers of size 128 and the activation

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}.$$

11.7 Loss function and training

The training target is $f_{\text{NN}}(\boldsymbol{\theta}(t)) \approx r(t)$, $r(t) = \dot{\boldsymbol{\theta}}_{\text{true}}(t) - f_{\text{physics}}(\boldsymbol{\theta}(t))$, with the mean-squared-error loss

$$\mathcal{L} = \frac{1}{TN} \sum_{t=1}^{T} \sum_{i=1}^{N} \left(r_i^{\text{true}}(t) - r_i^{\text{NN}}(t) \right)^2,$$

where T is the number of time samples, N the number of oscillators (system dimension), $r_i^{\text{true}}(t)$ the true physics residual and $r_i^{\text{NN}}(t)$ the neural-network prediction.

11.8 Optimization procedure

The optimization objective is

$$\min_{\theta_{\text{NN}}} \mathcal{L}(\theta_{\text{NN}}) = \frac{1}{TN} \sum_{t=1}^{T} \sum_{i=1}^{N} \left(r_i^{\text{true}}(t) - f_{\text{NN},i}(x(t); \theta_{\text{NN}}) \right)^2.$$

The training strategy is supervised learning on the physics residuals with mini-batch gradient descent (batch size = 64). The Adam update is

$$\theta_{k+1} = \theta_k - \alpha \frac{\hat{m}_k}{\sqrt{\hat{v}_k} + \varepsilon}, \qquad \alpha = 10^{-3},$$

where $\hat{m}_k$ is the first moment (mean gradient) and $\hat{v}_k$ the second moment (gradient variance). *Early stopping:* the validation loss is monitored each epoch, training stops after 25 epochs without improvement, and the best model parameters are saved. *Interpretation:* the optimization minimizes only the physics modeling error, preserving the physical structure.

11.9 Evaluation and physical interpretation

Hybrid time integration.

$$\boldsymbol{\theta}(t + \Delta t) = \boldsymbol{\theta}(t) + \left(f_{\text{physics}}(\boldsymbol{\theta}(t)) + f_{\text{NN}}(\boldsymbol{\theta}(t)) \right) \Delta t.$$

Order parameter (synchronization).

$$r(t) = \left| \frac{1}{N} \sum_{j=1}^{N} e^{\mathrm{i}\theta_j(t)} \right|.$$

The hybrid model reduces the MSE with respect to the physics-only model, improves long-term rollout stability and preserves physical interpretability.

11.10 Summary of the correction scheme

The classical Kuramoto ODE cannot fully capture nonlinearities, patient-specific effects or noise in real tremor data. The solution is to add a neural-network correction, $\dot{\theta}_i^{\text{pred}} = \dot{\theta}_i^{\text{Kuramoto}} + f_{\text{NN}}(\boldsymbol{\theta})$, in five steps: (1) compute the real phase derivative from data, $\dot{\theta}_i^{\text{data}}$; (2) compute the Kuramoto prediction $\dot{\theta}_i^{\text{Kuramoto}}$; (3) calculate the residual $r_i(t) = \dot{\theta}_i^{\text{data}} - \dot{\theta}_i^{\text{Kuramoto}}$; (4) train the network f_{NN} to learn the residual; (5) correct the model.

12 Results of the Hybrid Model

12.1 Neural-network training performance

Table 6: Hybrid model: train/validation loss over epochs.

Epoch	Train	Validation
1	4949.582	5006.061
10	4749.027	4815.677
20	4624.698	4715.501
30	4597.695	4689.726
40	4560.322	4672.660
50	4514.790	4647.444
60	4472.422	4611.189
70	4452.348	4593.376
80	4418.022	4584.276
90	4389.138	4577.872
100	4364.020	4592.153
120	4318.204	4566.635

Early stopping occurred at epoch 161; the best validation loss is 4552.973.

12.2 One-step residual prediction

Objective: compare predicted and true residuals one step ahead. The dataset has 4150 samples, 15 input features (sin, cos and ω for 5 oscillators) and 5 targets (data − Kuramoto residuals). The **one-step MSE** is 4314.358. The estimated natural frequencies are

$$\boldsymbol{\omega} = [23.01,\ 8.14,\ 35.92,\ 37.13,\ 25.57].$$

Interpretation: the network successfully learns corrections for each oscillator (input: $\boldsymbol{\theta}(t)$, $\sin\boldsymbol{\theta}$, $\cos\boldsymbol{\theta}$; output: residuals).

12.3 Multi-step hybrid rollout

The hybrid update equation is

$$\theta_i^{\text{pred}}(t+dt) = \theta_i^{\text{pred}}(t) + dt\Big[\underbrace{\omega_i + K\sum_{j=1}^{5}\sin(\theta_j - \theta_i)}_{\text{Kuramoto}} + \underbrace{f_{\text{NN}}(\boldsymbol{\theta})}_{\text{NN correction}}\Big].$$

The performance metrics are the multi-step MSE (angles) 3.0115 and the final order parameter (true = 0.4247, predicted = 0.4203). *Interpretation:* the low MSE indicates accurate multi-step prediction; the close order parameter indicates that synchronization patterns are preserved; the network effectively corrects the Kuramoto model for patient-specific tremor dynamics.

12.4 Summary and key takeaways

The objective was to improve the classical Kuramoto model for Parkinson's tremor dynamics. The steps performed were: (1) extraction of the phase trajectories $\theta_i(t)$ from tremor signals using the Hilbert transform; (2) estimation of the intrinsic frequencies ω_i and the coupling K from data; (3) computation of the residual dynamics $r_i(t) = \dot{\theta}_i^{\text{data}} - \dot{\theta}_i^{\text{Kuramoto}}$; (4) training of the network $f_{\text{NN}}(\boldsymbol{\theta})$ to learn the residuals (physics-informed approach); (5) evaluation: one-step prediction MSE 4314.36, multi-step rollout MSE (angles) 3.01, order parameter true = 0.4247 and predicted = 0.4203.

12.5 Visual representation

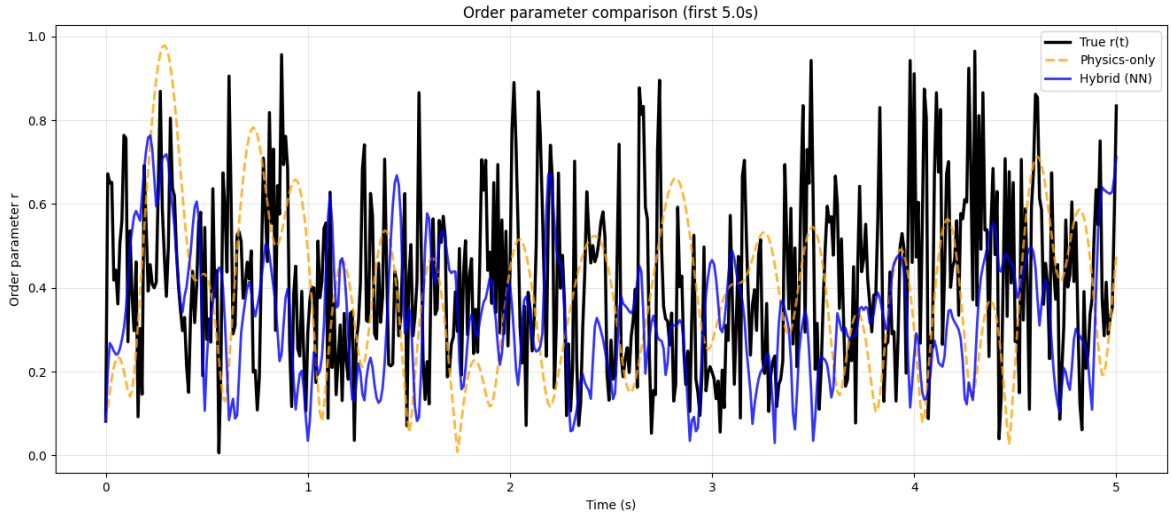


Figure 6: Order-parameter comparison over the first 5 s: true $r(t)$, physics-only model and hybrid (NN) model.

12.6 Quantitative comparison

Table 7: Performance: physics-only vs. hybrid model.

Metric	Physics	Hybrid	Improvement
One-step derivative prediction			
MSE	5.089×10^3	4.314×10^3	+15.22%
Multi-step rollout (41.5 s)			
Angular MSE	3.294	3.012	+8.57%
Order parameter (final time), true value $r_{\text{data}} = 0.4247$			
Predicted	0.1641	0.4203	—
Relative error	61.4%	1.0%	98.3% reduction

Key finding: the pure Kuramoto model predicts $r = 0.164$ against the true $r = 0.425$, i.e. it fundamentally mischaracterizes PD tremor synchronization.

12.7 Key observations

1. **Short-term accuracy.** The hybrid model improves one-step prediction by 15.22%; the network successfully corrects systematic deviations from idealized physics.
2. **Long-term stability.** The 8.57% improvement is maintained over 4,150 time steps (41.5 s); the residual corrections remain bounded, so there is no error accumulation—a critical advantage over purely data-driven models, which often diverge.
3. **Synchronization fidelity.** Physics-only: 61.4% error in the order parameter; hybrid: 1.0% error (a 60-fold reduction). *Conclusion:* the classical Kuramoto coupling is inadequate for PD tremor.

Part III

Advanced Hybrid Model: Physics, Residual Network and Coupling Network

13 Physics-Informed Neural Augmentation of the Kuramoto Model

13.1 Formulation

The classical Kuramoto model is

$$\frac{\mathrm{d}\theta_i}{\mathrm{d}t} = \omega_i + \frac{K}{N}\sum_{j=1}^{N}\sin(\theta_j - \theta_i).$$

We augment the dynamics using a physics-informed hybrid formulation:

$$\frac{\mathrm{d}\theta_i}{\mathrm{d}t} = \omega_i + \frac{K + \Delta K(\boldsymbol{\theta})}{N}\sum_{j=1}^{N}\sin(\theta_j - \theta_i) + r_i(\boldsymbol{\theta}), \tag{11}$$

where θ_i is the phase of oscillator i, $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_N)$ the system state, ω_i the intrinsic frequency, K the nominal coupling (physics prior), $\Delta K(\boldsymbol{\theta})$ the NN-learned coupling correction and $r_i(\boldsymbol{\theta})$ the NN residual (unmodeled dynamics).

13.2 Modeling approach

Parkinsonian tremor dynamics can be approximated as a network of interacting phase oscillators with partially unknown dynamics.

- **Phase representation.** Multivariate tremor signals are transformed into phase trajectories $\boldsymbol{\theta}(t) = (\theta_1(t), \ldots, \theta_N(t))$ using PCA and the Hilbert transform.
- **Physics-based prior.** Oscillator interactions are governed by a Kuramoto-type coupling; natural frequencies ω_i are estimated directly from data.
- **Learning unmodeled dynamics.** Neural networks are introduced to capture deviations from the idealized Kuramoto dynamics while preserving the physical structure.

13.3 Objective

1. **Extract phase trajectories:** transform multivariate tremor signals into phase variables $\boldsymbol{\theta}(t)$ using PCA and the Hilbert transform.
2. **Estimate intrinsic frequencies:** compute the natural frequencies ω_i directly from phase derivatives.
3. **Establish a physics-based baseline:** simulate classical Kuramoto dynamics with fixed coupling K.
4. **Learn residual dynamics:** train a residual neural network to model deviations from the Kuramoto dynamics.
5. **Identify state-dependent coupling effects:** learn a neural correction $\Delta K(\boldsymbol{\theta})$ while freezing the residual model.

6. **Evaluate predictive and dynamical accuracy:** compare the physics-only, hybrid and advanced models and assess long-term synchronization via the order parameter.

13.4 Why add a neural network?

Motivation. The classical Kuramoto model provides a strong physics-based prior, but it cannot fully capture the complexity of Parkinsonian tremor dynamics. We add (i) a **residual neural network** to model unobserved and non-ideal dynamics and (ii) a **coupling neural network** to learn the state-dependent interaction strength. The residual network is added directly to the phase dynamics, $\dot{\theta}_i = f_{\text{Kuramoto}}(\boldsymbol{\theta}) + r_i(\boldsymbol{\theta})$, while the coupling network modifies the interaction strength, $K \to K + \Delta K(\boldsymbol{\theta})$.

14 Methodology of the Advanced Hybrid Model

14.1 From raw data to phase trajectories

Goal. Transform raw multivariate tremor recordings into a reduced set of phase trajectories suitable for Kuramoto modeling.

- **Raw data:** the ALAMEDA Parkinson's tremor dataset with 4151 time samples and 97 sensor-derived features; the measurements are multivariate, noisy and non-oscillatory.
- **Dimensionality reduction:** principal component analysis (PCA) is applied to the raw signals, $\mathbf{s}(t) \to (x_1(t), \ldots, x_N(t))$, retaining $N = 5$ dominant components.
- **Phase construction:** each PCA component is transformed via the Hilbert transform into the analytic signal $z_i(t) = x_i(t) + \mathrm{i}\,\mathcal{H}[x_i(t)]$; the instantaneous phase is $\theta_i(t) = \arg(z_i(t))$, wrapped to $[-\pi, \pi)$.

14.2 Estimation of natural frequencies and coupling

Goal. Separate the intrinsic oscillator properties from the interaction effects.

- **Natural frequency estimation.** The phase trajectories are first unwrapped to ensure continuity, and the natural frequencies are estimated as time-averaged phase velocities, $\omega_i \approx \text{median}_t\left(\mathrm{d}\theta_i(t)/\mathrm{d}t\right)$.
- **Coupling modeling strategy.** A nominal coupling constant K is fixed as a physics-based prior, and a neural network learns a state-dependent correction, $K \to K + \Delta K(\boldsymbol{\theta})$.
- **Key design choice.** Natural frequencies are estimated directly from data; the coupling variability is learned by a neural network without breaking the Kuramoto structure.

14.3 Why physics alone is insufficient

The physics-based baseline is $\dot{\theta}_i^{\text{phys}} = \omega_i + \frac{K}{N}\sum_{j=1}^{N}\sin(\theta_j - \theta_i)$, whereas the observed phase dynamics from data satisfy $\dot{\theta}_i^{\text{data}} \neq \dot{\theta}_i^{\text{phys}}$. The sources of the discrepancy are unmodeled nonlinear interactions, non-stationary or state-dependent coupling, and measurement noise and real-world variability. The key insight is

$$\dot{\theta}^{\text{data}} = f_{\text{phys}}(\boldsymbol{\theta}) + \text{unmodeled dynamics}.$$

14.4 Hybrid model: residual neural network

The phase dynamics are decomposed as

$$\dot{\boldsymbol{\theta}} = f_{\text{phys}}(\boldsymbol{\theta}) + r_\theta(\boldsymbol{\theta}),$$

where $f_{\text{phys}}(\boldsymbol{\theta})$ is the Kuramoto physics model and $r_\theta(\boldsymbol{\theta})$ a neural residual correction. The residual learning objective is $r_\theta(\boldsymbol{\theta}) \approx \dot{\boldsymbol{\theta}}^{\text{data}} - f_{\text{phys}}(\boldsymbol{\theta})$, with the training loss

$$\mathcal{L}_{\text{res}} = \left\| r_\theta(\boldsymbol{\theta}) - \left(\dot{\boldsymbol{\theta}}^{\text{data}} - f_{\text{phys}}(\boldsymbol{\theta})\right) \right\|^2.$$

The neural network learns only the missing physics.

14.5 Residual neural-network training

Goal: learn the dynamics not explained by the Kuramoto model. The physics baseline is $f_{\text{phys},i}(\boldsymbol{\theta}) = \omega_i + \frac{K}{N}\sum_{j=1}^{N}\sin(\theta_j - \theta_i)$, the residual target is $y_{\text{res}}(\boldsymbol{\theta}) = \dot{\boldsymbol{\theta}}^{\text{data}} - f_{\text{phys}}(\boldsymbol{\theta})$, and the residual network is $r_\theta : \mathbb{T}^N \to \mathbb{R}^N$, $r_\theta(\boldsymbol{\theta}) \approx y_{\text{res}}(\boldsymbol{\theta})$. The loss is

$$\mathcal{L}_{\text{res}} = \frac{1}{TN}\sum_{k,i}\Big(r_{\theta,i}(\boldsymbol{\theta}(t_k)) - \left(\dot{\theta}_{i,\text{data}} - f_{\text{phys},i}(\boldsymbol{\theta})\right)\Big)^2,$$

and the optimization step is

$$\theta_{\text{res}}^{(m+1)} = \theta_{\text{res}}^{(m)} - \eta\,\nabla_{\theta_{\text{res}}}\mathcal{L}_{\text{res}}.$$

14.6 Freezing the residual neural network

After the residual training, $r_\theta(\boldsymbol{\theta}) \approx \dot{\boldsymbol{\theta}}_{\text{data}} - f_{\text{phys}}(\boldsymbol{\theta})$. The network is frozen in code by `p.requires_grad = False`, which has the mathematical implication

$$\nabla_{\theta_{\text{res}}}\mathcal{L} = 0 \quad \Rightarrow \quad r_\theta(\boldsymbol{\theta}) \text{ is fixed.}$$

Freezing is necessary because it

- prevents the coupling network from modifying the residual dynamics,
- enforces a separation of roles: *unstructured effects* $\neq$ *structured coupling*,
- ensures the identifiability of $\Delta K(\boldsymbol{\theta})$.

Interpretation:

$$\dot{\boldsymbol{\theta}} = f_{\text{phys}}(\boldsymbol{\theta}) + \underbrace{r_\theta(\boldsymbol{\theta})}_{\text{explained}} + \underbrace{f_{\text{coup}}(\boldsymbol{\theta}, \Delta K)}_{\text{to be learned}}.$$

At this stage the unstructured dynamics are considered explained.

14.7 Coupling neural network: state-dependent interaction

After freezing the residual network, a second neural component is introduced to model *structured coupling corrections*. The coupling decomposition is $K \longrightarrow K + \Delta K(\boldsymbol{\theta})$, where K is the nominal physics-based coupling and $\Delta K(\boldsymbol{\theta})$ the neural correction (scalar-valued). The coupling network is a map $\Delta K : \mathbb{T}^N \to \mathbb{R}$. Its key properties are that it outputs a single scalar per system state, preserves the Kuramoto interaction structure, and cannot override the residual dynamics.

14.8 Resulting dynamics of the advanced hybrid model

After incorporating both neural components, the phase dynamics take the form

$$\dot{\boldsymbol{\theta}} = f_{\text{phys}}(\boldsymbol{\theta}; K) + r_\theta(\boldsymbol{\theta}) + f_{\text{coup}}\big(\boldsymbol{\theta}; \Delta K(\boldsymbol{\theta})\big),$$

where $f_{\text{phys}}(\boldsymbol{\theta}; K)$ is the classical Kuramoto dynamics, $r_\theta(\boldsymbol{\theta})$ the residual neural correction (unstructured effects) and $f_{\text{coup}}(\boldsymbol{\theta}; \Delta K(\boldsymbol{\theta}))$ the structured coupling correction. All unknown dynamics are thus decomposed into interpretable components, and the physical structure is preserved by construction.

14.9 Coupling-network training objective

Goal: learn the state-dependent coupling after the residual dynamics are explained. The training target is

$$y_{\text{coup}}(\boldsymbol{\theta}) = \dot{\boldsymbol{\theta}}_{\text{data}} - \big(f_{\text{phys}}(\boldsymbol{\theta}; K) + r_\theta(\boldsymbol{\theta})\big),$$

the coupling-induced dynamics are

$$f_{\text{coup},i}(\boldsymbol{\theta}) = \frac{\Delta K(\boldsymbol{\theta})}{N} \sum_{j=1}^{N} \sin(\theta_j - \theta_i),$$

and the loss is $\mathcal{L}_{\text{coup}} = \|f_{\text{coup}}(\boldsymbol{\theta}) - y_{\text{coup}}(\boldsymbol{\theta})\|^2$, under the constraint that the residual network is frozen ($\nabla_{\theta_{\text{res}}} \mathcal{L} = 0$).

14.10 Optimization of the coupling network

The objective is

$$\min_{\theta_{\text{coup}}} \ \mathcal{L}_{\text{coup}} = \Big\| f_{\text{coup}}\big(\boldsymbol{\theta}; \Delta K(\boldsymbol{\theta}; \theta_{\text{coup}})\big) - y_{\text{coup}}(\boldsymbol{\theta}) \Big\|_2^2,$$

where $\|\cdot\|_2^2$ denotes the squared Euclidean norm (MSE), averaged over all oscillators. The update rule (Adam) is

$$\theta_{\text{coup}}^{(t+1)} = \text{Adam}\left(\theta_{\text{coup}}^{(t)}, \nabla_{\theta_{\text{coup}}} \mathcal{L}_{\text{coup}};\ \alpha = 10^{-3}\right),$$

with $\alpha = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$; the physics and residual networks are frozen.

14.11 Neural-network architectures

Both networks take $\boldsymbol{\theta}(t) \in \mathbb{R}^N$ as input ($N = 5$ in the first experiment).

- **Residual neural network:** learns the unmodeled phase dynamics; fully connected feedforward network; layers $N \to 64 \to 64 \to N$; activation Tanh; output $r_\theta(\boldsymbol{\theta}) \in \mathbb{R}^N$.
- **Coupling neural network:** learns the state-dependent coupling correction; fully connected feedforward network; layers $N \to 32 \to 1$; activation Tanh; output $\Delta K(\boldsymbol{\theta}) \in \mathbb{R}$.

15 Results of the Advanced Hybrid Model

15.1 Experiment 1 ($N = 5$)

Residual network training (hybrid model). Training setup: 200 epochs, mean-squared-error loss, Adam optimizer, CPU.

Table 8: Experiment 1: residual-network training (hybrid model).

Epoch	Train	Validation
1	5051.44	5174.62
20	4711.73	4821.54
60	4582.40	4738.16
120	4456.80	4685.55
200	4301.98	4639.99

Observations: monotonic decrease of the training and validation loss, no signs of overfitting, best validation loss 4637.21; the residual network successfully captures unmodeled tremor dynamics.

Coupling network training (advanced model). Training setup: 200 epochs, MSE loss, Adam optimizer, residual network **frozen**, CPU.

Table 9: Experiment 1: coupling-network training (advanced model).

Epoch	Train	Validation
1	5163.72	5242.39
20	5161.70	5237.83
60	5155.94	5231.02
120	5151.60	5225.67
200	5146.26	5223.43

Observations: stable but slow convergence due to the constrained parameter space; the optimization is limited to structured coupling terms; no overfitting is observed; best validation loss 5223.23.

Model evaluation: MSE comparison. Mean squared error on test data:

Table 10: Experiment 1: test MSE of the three models.

Model	Test MSE	Improvement
Physics-only	4958.90	—
Hybrid (physics + residual NN)	4360.99	12.06% vs. physics-only
Advanced (physics + residual NN + coupling NN)	4344.90	0.37% vs. hybrid

Observations: the residual network significantly improves the model performance by capturing unmodeled dynamics; the coupling network provides a small additional gain, indicating that most dynamics are already captured by the hybrid model; the physics-based model alone is insufficient for accurately modeling Parkinson's tremor dynamics; and the decomposition into physics + residual + coupling allows interpretable and structured learning.

15.2 Experiment 2 ($N = 25$)

Architectures. Both networks take $\boldsymbol{\theta}(t) \in \mathbb{R}^{25}$ as input. The residual network is a fully connected feedforward network with layers $25 \to 64 \to 64 \to 25$, Tanh activation and output $r_\theta(\boldsymbol{\theta}) \in \mathbb{R}^{25}$; the coupling network has layers $25 \to 32 \to 1$, Tanh activation and output $\Delta K(\boldsymbol{\theta}) \in \mathbb{R}$.

Residual network training (hybrid model). Training setup: 150 epochs, MSE loss, Adam optimizer, CPU.

Table 11: Experiment 2: residual-network training (hybrid model).

Epoch	Train	Validation
1	5398.95	5462.64
20	5149.58	5229.63
40	5026.06	5224.27

Early stopping at epoch 53 with best validation loss 5215.30. Observations: monotonic decrease of the training and validation loss, early stopping applied to prevent overfitting, and the residual network captures unmodeled tremor dynamics.

Coupling network training (advanced model). Training setup: 400 epochs, MSE loss, Adam optimizer, residual network **frozen**, CPU.

Table 12: Experiment 2: coupling-network training (advanced model).

Epoch	Train	Validation
1	6178.16	6245.90
20	6170.02	6249.32

Early stopping at epoch 21 with best validation loss 6245.90. Observations: stable but slow convergence due to the constrained parameter space, optimization limited to structured coupling terms, no overfitting, and the coupling network provides a fine-tuning correction.

Table 13: Experiment 2: test MSE of the three models.

Model	Test MSE	Improvement
Physics-only	5652.94	—
Hybrid (physics + residual NN)	5224.96	7.57% vs. physics-only
Advanced (physics + residual NN + coupling NN)	5218.39	0.13% vs. hybrid

Model evaluation: MSE comparison. Observations: the residual network significantly improves the model performance; the coupling network provides a small additional gain; the physics-only model is insufficient for accurately modeling Parkinson's tremor dynamics; the decomposition into physics + residual + coupling allows interpretable and structured learning.

16 Conclusion

This report followed one line of work in three stages and built, step by step, a physics-informed neural description of coupled oscillators.

Part I — surrogate. The Kuramoto system is a first-order, nonlinear, homogeneous system with constant coefficients whose vector field is continuous and Lipschitz, so a solution exists (Peano) and is unique (Osgood). A neural surrogate of the flow map Φ_T, working in the circular $(\cos, \sin)$ embedding and trained with a physics-informed loss (MSE, order parameter, phase variance), predicts oscillator phases with a mean angular error of 17.16° (Experiment B) and 19.42° (Experiment A), while the order parameter is preserved with $\mathrm{MAE}_r = 0.0080$ and 0.00264, respectively. A single forward pass replaces numerical integration; large angular errors are rare and correspond to isolated outliers.

Part II — hybrid model. Applied to Parkinsonian tremor recordings, the classical Kuramoto model is not sufficient: it predicts a final order parameter of 0.164 against the true 0.425. A residual neural network learns the mismatch between data and physics and yields a 15.22% lower one-step MSE, an 8.57% lower multi-step angular MSE that is maintained over 4,150 time steps (41.5 s) without error accumulation, and a reduction of the order-parameter error from 61.4% to 1.0%.

Part III — advanced hybrid model. Adding a scalar, state-dependent coupling correction $\Delta K(\boldsymbol{\theta})$ on top of a *frozen* residual network gives a decomposition *physics + residual + coupling* in which each component has a distinct role. The residual network produces the main gain (12.06% for $N = 5$ and 7.57% for $N = 25$ over pure physics), whereas the coupling network provides only a small additional improvement (0.37% and 0.13% over the hybrid model), which indicates that most dynamics are already captured by the hybrid model.

Table 14: Key quantitative results of the three stages.

Stage	Setting	Main result
I. Surrogate	Exp. A / Exp. B ($N = 10$ oscillators)	MAE_θ = 19.42° / 17.16°; $\mathrm{MAE}_r = 0.00264$ / 0.0080
II. Hybrid vs. physics	ALAMEDA PD, 5 oscillators	one-step MSE +15.22%; multi-step MSE +8.57%; r-error 61.4% → 1.0%
III. Hybrid vs. physics	$N = 5$ / $N = 25$	test MSE +12.06% / +7.57%
III. Advanced vs. hybrid	$N = 5$ / $N = 25$	test MSE +0.37% / +0.13%

Outlook (author's remarks). The observations suggest several directions: reducing the rare large angular errors of the surrogate; richer coupling structures than a single scalar $\Delta K(\boldsymbol{\theta})$, since the scalar correction yields only a marginal gain and converges slowly because of its constrained parameter space; patient-specific parameters; and linking the learned phase–amplitude description with closed-loop DBS strategies discussed in Section 9.3.

A Project Evaluation Summaries

The three projects were accompanied by a short evaluation summary (the third one titled "strict and objective"). The scores are reproduced in Table 15.

Table 15: Project evaluation summaries (scores out of 10).

Criterion	Project #1 (Surrogate)	Project #2 (Hybrid)	Project #3 (Advanced hybrid)
Problem formulation and clarity	9	10	10
Baseline method	8.5	10	10
Method justification (and suitability / depth)	8	9	9
Experimental design and correctness	9	9	9
Quality of analysis and insight	9	9	9

Comments on Project #1: the problem is clearly formulated as surrogate modeling of the Kuramoto flow map $\Phi_T : \boldsymbol{\theta}_0 \mapsto \boldsymbol{\theta}(T)$; high-accuracy numerical integration with Dormand–Prince RK45 is an appropriate and reliable baseline; the surrogate architecture and the circular embedding are well justified; dataset generation, the training/validation/test split and the optimization strategy are sound; and the results are interpreted with a clear distinction between local phase accuracy and global synchronization behavior. In Project #2 the baseline method is the Kuramoto model.

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